

A Deep Learning-Based Portfolio Risk Forecasting Application

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Abstract

This report presents a robust and interactive Streamlit application designed for forecasting the Value at Risk (VaR) of stock portfolios using advanced deep learning models. The application strategically employs two distinct deep learning methodologies: **DeepAR**, a univariate probabilistic forecasting method from Amazon's GluonTS library, to model asset-level return distributions for individual stocks, and **DeepVAR**, a multivariate approach for portfolio-level risk analysis to capture the interconnectedness of multiple stocks. This report contrasts both in terms of modeling assumptions, structure, and use cases. The objective is to provide a detailed, reproducible, and academically rigorous reference for developing or extending AI-based financial risk tools. The application provides an intuitive user interface for portfolio construction, risk parameter adjustment, and real-time visualization of risk metrics. This tool aims to bridge the gap between theoretical risk management and practical application, offering a user-friendly platform for understanding and mitigating financial risks.

Introduction and Motivation

In today's fast-moving and uncertain financial landscape, effective portfolio risk management is crucial for both institutional and individual investors. Traditional VaR methods like Historical Simulation and Parametric VaR often fall short due to rigid distributional assumptions or an inability to account for dynamic temporal relationships. The increasing availability of financial data and advancements in computational power have paved the way for more sophisticated, data-driven approaches to risk management. Deep learning, with its ability to model complex patterns and dependencies in time series data, has emerged as a promising tool in this domain. This project explores the application of deep learning techniques, specifically leveraging DeepAR for individual asset forecasting and DeepVAR for comprehensive portfolio risk assessment, to forecast portfolio risk and provide a more accurate and adaptable risk assessment tool.

Traditional risk management techniques often struggle to capture the complexities of modern financial markets. Parametric methods, which rely on assumptions about the distribution of asset returns (e.g., normality), can be unreliable when these assumptions are violated, as is often the case in real-world financial data. Non-parametric methods, such as Historical Simulation, are limited by their reliance on past data and their inability to adapt to changing market conditions. Deep learning models, on the other hand, can learn complex, non-linear relationships from historical data and adapt to changing market dynamics. This makes them well-suited for forecasting risk in dynamic and uncertain environments.

To overcome these limitations, DeepVaR uses DeepAR for forecasting individual stock returns; a neural time series model that can learn complex seasonality and trends from historical data while producing full predictive distributions rather than point estimates. DeepVAR is then applied at the portfolio level to capture the interdependencies between assets, offering a holistic risk view. The app forecasts daily return distributions and calculates risk metrics such as VaR across multiple horizons and confidence levels. By modelling the full distribution of potential returns, DeepAR and DeepVAR provide a more comprehensive view of risk than traditional methods that focus solely on point estimates. The ability to generate probabilistic forecasts allows for the calculation of VaR at various confidence levels, providing a more nuanced understanding of potential losses.

Objective and Scope

The main objectives of this application and report are to:

- Provide an end-to-end risk analysis pipeline using modern deep learning techniques, leveraging DeepAR for individual assets and DeepVAR for portfolio-level analysis.
- Allow users to input portfolio weights and observe VaR forecasts interactively.
- Compare and contrast DeepAR (used in the app for individual assets) with DeepVAR (used for portfolio risk), shedding light on their appropriate use cases.
- Encourage further research by offering an extendable codebase and detailed documentation.

The scope of this project includes the development of a Streamlit application that allows users to construct portfolios, train DeepAR models on historical stock data for individual assets, and DeepVAR models to assess overall portfolio risk, and visualize VaR forecasts.

Value at Risk

Value at Risk (VaR) measures the maximum expected loss over a time period at a given confidence level. For example, a 10-day 99% VaR of \$500 means there's a 1% chance the portfolio will lose more than \$500 in the next 10 days. VaR is a widely used risk measure in the financial industry, providing a quantitative estimate of potential losses. It is used by portfolio managers, risk officers, and regulators to assess and manage financial risk. The interpretation of VaR is straightforward: it provides a threshold below which losses are expected to fall with a certain probability. This allows stakeholders to understand the potential downside risk of an investment portfolio or financial institution.

System Overview and Workflow

Architecture Components

- Frontend: Built with Streamlit for a clean UI.
- Backend: DeepAR model from GluonTS with MXNet backend for individual stock forecasting, and DeepVAR for portfolio analysis.
- Data Pipeline: Ingests CSVs of historical prices, computes log returns.
- Risk Engine: Forecasts returns (DeepAR and DeepVAR), calculates VaR, visualizes outputs.

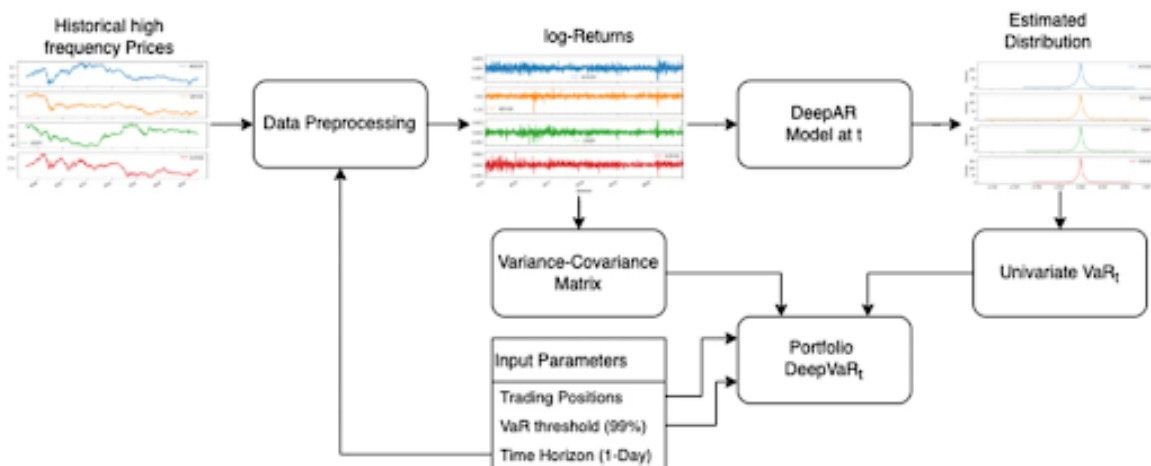


Figure 3: Risk Forecasting Pipeline. From historical high-frequency prices, the system preprocesses data into log-returns, trains a DeepAR model per asset, and estimates univariate distributions. These are then aggregated using a portfolio-level DeepVAR model to compute Value at Risk (VaR) based on user-defined input parameters.

The system architecture is designed to be modular and scalable, with distinct components for the frontend, backend, data pipeline, and risk engine. The Streamlit frontend provides a user-friendly interface for portfolio construction and risk analysis. The backend, powered by the DeepAR model from the GluonTS library for individual stock forecasting and DeepVAR for portfolio-level risk assessment, handles the forecasting of asset returns. The data pipeline ingests historical stock price data, preprocesses it, and transforms it into a format suitable for model training. The risk engine calculates VaR based on the forecasted return distributions and provides visualizations of the results. This modular architecture allows for easy modification and extension of the system. For example, the backend can be replaced with a different forecasting model, or the data pipeline can be modified to ingest data from different sources.

User Flow

- User Input: Select stocks and assign weights.
- Data Prep: Align, clean, and transform prices into returns.
- Model Training: Fit DeepAR model on each asset's returns.
- Forecasting: Predict 10-day ahead return distributions using DeepAR for individual assets and DeepVAR for portfolio-level dependencies.
- Risk Analysis: Compute VaR for each horizon using DeepVAR outputs.
- Visualization: Display forecast intervals, VaR bands, and PnL charts.

The user flow is designed to be intuitive and straightforward, guiding the user through the process of portfolio construction, model training, risk analysis, and visualization. The user begins by selecting stocks and assigning weights to create a portfolio. The system then preprocesses the historical price data for these stocks, aligning dates, cleaning missing values, and computing returns. The DeepAR model is trained on the historical return data for each asset. Once the model is trained, it is used to forecast future return distributions for each asset. DeepVAR is then applied to these individual forecasts to capture portfolio-level dependencies. These forecasts are then used to calculate VaR for the portfolio at different confidence levels and time horizons. The results are visualized using interactive charts and graphs, providing the user with a clear understanding of the portfolio's risk profile. The user can adjust the portfolio weights and risk parameters to explore different scenarios and understand the impact on VaR.

Data Engineering and Preprocessing

Input Data

The input data consists of historical stock price data in CSV format. Each CSV file should contain two columns: a datetime column indicating the date and time of the price observation, and a price column representing the adjusted close price of the stock. The data should be at daily frequency, meaning that there is one price observation per day. The adjusted close price is used to account for stock splits and dividends, providing a more accurate representation of the stock's historical performance. The data should be clean and accurate, with minimal missing values or errors. The quality of the input data is critical for the accuracy of the VaR forecasts.

Cleaning

- Dates are parsed and set as index.
- Missing values are forward-filled and then dropped if persistent.

- Stocks are aligned to the intersection of all trading dates.

The data cleaning process involves several steps to ensure that the data is in a suitable format for model training. First, the datetime column is parsed and set as the index of the data frame. This allows for easy time series manipulation and analysis. Next, missing values are handled using a forward-fill approach, where missing values are replaced with the previous available value. If missing values persist after forward-filling, they are dropped from the data frame. Finally, the stock data is aligned to the intersection of all trading dates. This ensures that all stocks have data for the same period, which is necessary for portfolio construction and risk analysis. The data cleaning process is crucial for ensuring the accuracy and reliability of the VaR forecasts.

Return Computation

The returns are computed as the percentage change in price from one day to the next. This is a common way to represent the performance of a stock or portfolio. The returns are used as the target variable for the DeepAR model, which learns to predict future returns based on historical return data. The choice of return calculation method can impact the performance of the model. Other methods, such as log returns, may be more appropriate in some cases. The returns should be calculated consistently across all stocks in the portfolio to ensure accurate portfolio-level risk analysis.

Parametric, Semi-Parametric, and Non-Parametric Risk Modelling

Overview of Modeling Paradigms

Risk forecasting methods are often categorized based on the degree of assumptions made about the data-generating process:

(a) Parametric Models	(b) Non-Parametric Models	(c) Semi-Parametric Models
Assume a fixed functional form (e.g., Normal, t-distribution).	Make no assumptions about the distribution.	Combine data-driven flexibility with some structured assumptions.
Parameters are estimated from data, and inference is derived under these assumptions.	Rely purely on empirical data and resampling techniques.	Often use parametric models for conditional mean/variance, but allow non-parametric residuals or forecast distributions.
Examples: Variance-Covariance method (VaR under Gaussian returns), GARCH models.	Examples: Historical VaR, Kernel Estimation (KDE), Simulation, Density	Examples: Filtered Historical Simulation (FHS), DeepAR and DeepVAR (as used in this project), Quantile Regression Forests
Pros: Efficient when assumptions are met; analytically tractable.	Pros: Flexible; no distributional assumptions.	Pros: Adaptable; balance structure and flexibility.
Cons: Fragile under misspecification; poor tail modeling.	Cons: Requires large samples; poor generalisation for unseen regimes.	Cons: Interpretation can be opaque; tuning-intensive.

Why DeepAR Is Semi-Parametric

DeepAR falls under the semi-parametric modeling framework:

- Parametric component:
 - The network architecture (e.g., LSTM layers, likelihood families like Gaussian or Student-t) imposes a structured model of dependence.
- Non-parametric component:
 - Forecasts are generated via Monte Carlo sampling, with empirical quantile estimation used for VaR. No explicit assumption is made about the joint return distribution of the portfolio.
- Cross-series independence:
 - Although DeepAR assumes conditional independence across series, the final portfolio return distribution is assembled non-parametrically via simulation.

This duality allows the model to learn rich, temporal dynamics in individual assets while using distribution-free aggregation for portfolio-level risk estimation. This makes DeepAR a flexible and powerful tool for risk forecasting.

DeepAR Model and Forecasting

Model Description

DeepAR uses autoregressive RNNs (typically LSTMs) to model the conditional distribution of a time series given its past. It learns across multiple related time series and is particularly strong in handling cold-starts and non-linear seasonalities. In this application, DeepAR is specifically used to forecast the returns of individual stocks. DeepAR is a probabilistic forecasting model that uses recurrent neural networks (RNNs) to model the conditional distribution of a time series. The model learns to predict future values of the time series based on its past values. DeepAR is particularly well-suited for forecasting multiple related time series, such as the returns of different stocks in a portfolio. The model can learn from the patterns and dependencies across these time series, improving the accuracy of the forecasts. DeepAR is also effective at handling cold-start problems, where there is limited historical data available for a particular time series. The model can leverage information from other related time series to make more accurate predictions.

Implementation Details

- Context length: 15 days
- Prediction horizon: 10 days
- RNN Layers: 2 LSTM layers
- Epochs: 5 (can be increased)
- Dropout: 10% to prevent overfitting

A longer context length allows the model to learn more complex patterns, but also increases the computational cost. The prediction horizon determines how far into the future the model predicts. The number of RNN layers determines the depth of the model. Deeper models can learn more complex relationships, but are also more prone to overfitting. The number of epochs determines how many times the model iterates over the training data. More epochs can lead to better performance, but also increase the risk of overfitting. The dropout rate is a regularization technique that helps prevent overfitting by randomly dropping out neurons during training. These implementation details can be tuned to optimize the performance of the model for a particular dataset and application.

The `DeepARModel` class wraps GluonTS training and prediction routines, outputting 1000 samples per prediction time point. This allows for the generation of a distribution of possible outcomes, which is then used to calculate VaR.

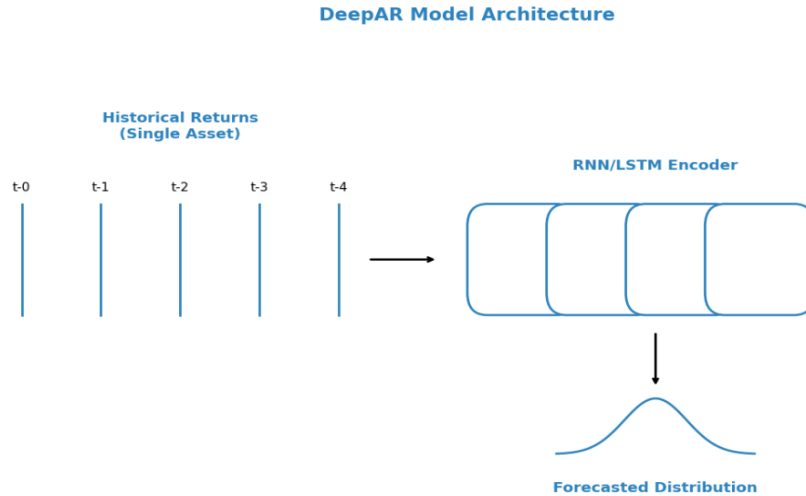


Figure 4: DeepAR Model Architecture. DeepAR models each asset's historical returns through an autoregressive RNN (LSTM) encoder, forecasting a full probabilistic distribution of future returns via Monte Carlo sampling.

Forecast Aggregation

After forecasting individual asset returns with DeepAR, we compute simulated portfolio returns using DeepVAR, which takes into account the interconnectedness. The forecasted returns for each asset are aggregated to compute the portfolio return. This is done by multiplying the forecasted return for each asset by its corresponding portfolio weight and summing across all assets. The resulting portfolio return represents the overall performance of the portfolio based on the model's forecasts. The VaR is then calculated from the distribution of portfolio returns, providing an estimate of the maximum expected loss at a given confidence level. The VaR is plotted with confidence bands to visualize the range of potential losses. This allows users to understand the uncertainty associated with the VaR estimate and make more informed risk management decisions.

DeepVAR vs. DeepAR: Conceptual and Practical Comparison

DeepVAR is a multivariate extension of DeepAR, designed to capture both temporal dependencies within each series and cross-series interactions (e.g., correlations). It combines vector autoregressive modeling with deep learning, typically using:

- A shared encoder-decoder RNN structure.
- Multivariate input and output streams.
- Cross-variable attention or multivariate LSTMs.

In this application, DeepVAR is specifically used for portfolio-level risk analysis, leveraging its ability to model the relationships between different assets. DeepVAR is a more sophisticated model than DeepAR, capable of capturing complex dependencies between multiple time series. It uses a shared encoder-decoder RNN structure to learn from the patterns and relationships across different time series. The multivariate input and output streams allow the model to handle multiple time series simultaneously. Cross-variable attention mechanisms or multivariate LSTMs can be used to capture the dependencies between different time series. DeepVAR is particularly well-suited for modeling financial markets, where the returns of different assets are often correlated. By capturing these correlations, DeepVAR can provide more accurate risk forecasts than DeepAR.

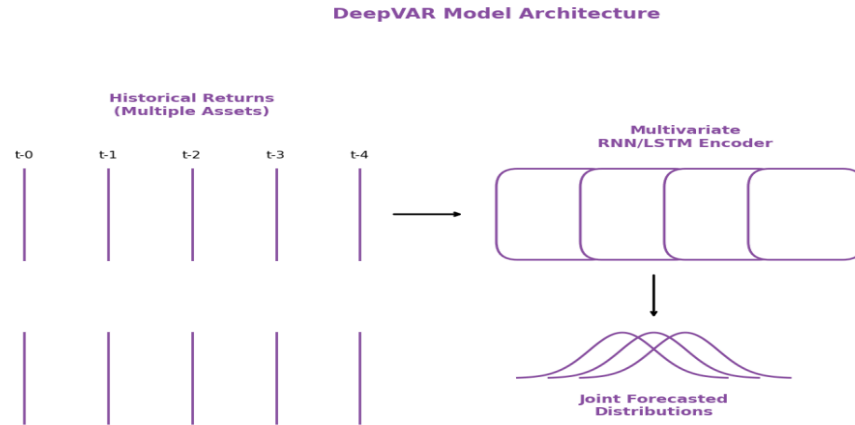


Figure 5: DeepVAR Model Architecture.

DeepVAR processes multiple series jointly using a multivariate RNN structure, capturing cross-asset dependencies.

Key Differences

Feature	DeepAR	DeepVAR
Input	Univariate series	Multivariate series
Cross-series dependence	Learns global patterns, no covariance	Learns explicit dependencies
Use case	Forecasting many series independently	Modeling co-movement and spillovers
Risk application	Individual asset VaR	Joint portfolio VaR, stress testing
Scalability	Scales well with many series	Computationally expensive for many series

How DeepAR and DeepVAR Connect

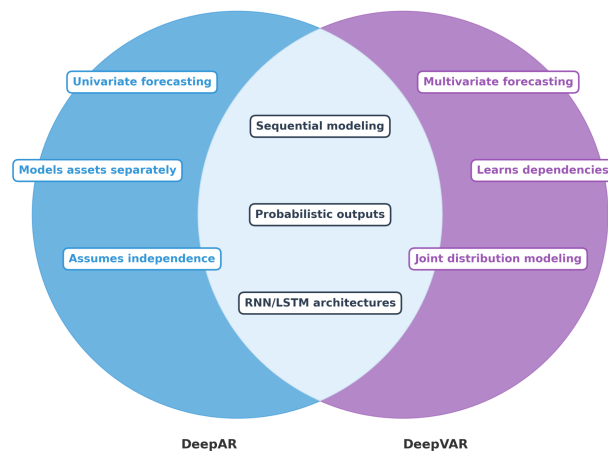


Figure 6: Comparative Overview of DeepAR and DeepVAR Models.

DeepAR independently forecasts individual time series assuming conditional independence, while DeepVAR jointly forecasts multiple series by capturing inter-series dependencies. Both models leverage sequential neural network architectures and probabilistic output generation.

In practice:

- DeepAR is ideal when each asset can be assumed conditionally independent given its past (or when series are weakly correlated). This is used as a first step to forecast each stock.
- DeepVAR is suited for closely connected assets (e.g., ETFs, sector stocks) where correlations and co-movements are crucial. This is used at the portfolio level to capture these interdependencies.

DeepAR and DeepVAR can be used in different scenarios depending on the assumptions about the relationships between the assets in the portfolio. If the assets are assumed to be conditionally independent given their past, DeepAR is a suitable choice. However, if the assets are closely connected and their correlations and co-movements are crucial, DeepVAR is a better choice. In practice, a hybrid approach may be used, where DeepAR is used for some assets and DeepVAR is used for others.

Future extensions of DeepVaR could include a switchable backend allowing users to choose DeepAR or DeepVAR depending on portfolio characteristics. This would provide users with greater flexibility and control over the risk modeling process.

Visual Analytics and Interpretation

Asset-Level Forecasts

Each asset's return forecast (generated by DeepAR) is shown as:

- Median forecast line.
- Shaded bands at 90% and 99% intervals (from DeepAR samples).

The asset-level forecasts are visualized using interactive charts that display the median forecast line and shaded bands representing the 90% and 99% confidence intervals. This allows users to understand the range of potential outcomes for each asset and the uncertainty associated with the forecasts. The median forecast line represents the most likely outcome, while the shaded bands represent the range of possible outcomes with a certain probability. The wider the shaded bands, the greater the uncertainty associated with the forecast.

Portfolio Risk Timeline

Displays:

- Mean expected portfolio return (PnL).
- 95% and 99% VaR lines.
- Optional cumulative PnL and drawdown estimates.

The portfolio risk timeline provides a comprehensive view of the portfolio's risk profile over time. It displays the mean expected portfolio return (PnL), the 95% and 99% VaR lines, and optional cumulative PnL and drawdown estimates. The mean expected portfolio return represents the average return that the portfolio is expected to generate over time. The VaR lines represent the maximum expected loss at a given confidence level. The cumulative PnL represents the total profit or loss generated by the portfolio over time. The drawdown estimates represent the maximum peak-to-trough decline in the portfolio's value over time. These visualizations allow users to understand the potential risks and rewards associated with the portfolio and make more informed investment decisions.

Allocation Breakdown

A dynamic pie chart visualizes the portfolio composition, helping users see exposure and diversification. This allows users to quickly assess the concentration of their portfolio and identify potential diversification opportunities.

Applications and Use Cases

- Risk Monitoring for Investors
- Financial Advisory and Education
- Research Prototyping

Limitations and Future Directions

Limitation	Planned Improvement
DeepAR assumes conditional independence	Current implementation uses DeepVAR for portfolio analysis to address this limitation
Static 10-day forecast horizon	Allow user-defined horizons
No economic covariates or news signals	Integrate macro and textual indicators
No backtesting module	Add VaR backtesting and performance diagnostics
Only historical data from CSVs	Add support for live financial data APIs

Conclusion

DeepVaR serves as a bridge between classical financial risk estimation and modern deep learning methods. By using DeepAR to generate probabilistic forecasts for portfolio assets, it enables intuitive, robust, and dynamic risk evaluation. The theoretical inclusion of DeepVAR in this report also lays the groundwork for future expansions that incorporate multivariate dependencies explicitly, enhancing realism and accuracy. The application provides a user-friendly platform for understanding and mitigating financial risks, making it a valuable tool for investors, financial advisors, and educators. The modular and extensible codebase allows for future enhancements and modifications, paving the way for further research and development in the field of AI-based financial risk management. As deep learning techniques continue to evolve, the DeepVaR application will remain a valuable tool for navigating the complexities of modern financial markets.

References

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